Unit 3 Progress Check: FRQ

Include correctly labeled diagrams, if useful or required, in explaining your answers. A correctly labeled diagram must have all axes and curves clearly labeled and must show directional changes. If the question prompts you to "Calculate," you must show how you arrived at your final answer.

Quantity of Output	Total Cost
0	$12
1	$14
2	$18
3	$24
4	$32
5	$42
6	$54
7	$68

The table above shows the total cost function for a typical firm producing hats in a perfectly competitive market. The market price for hats is $9 per hat.

(a) Calculate the average variable cost of the fifth unit. Show your work.

(b) What is the firm's profit-maximizing quantity of hats? Explain using marginal analysis.

(c) If the rent of the building the firm occupies increases, what will happen to the firm's profit-maximizing quantity of hats in the short run? Explain.

(d) Draw a correctly labeled graph showing the firm's demand and marginal cost curves, and show the profit-maximizing quantity of hats, labeled Q*.

1. Respond to all parts of the question.

Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response must show appropriate work in order to receive the point.

0	1 ✓

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The response calculates the average variable cost of the fifth unit and shows appropriate work.

$\text{TVC of 5 units} = TC - TFC = \$42 - \$12 = \$30.\ \text{AVC of 5 units} = \text{TVC/Q} = \$30/5 = \$6$

Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response must both identify the profit-maximizing output as 4 hats and explain why using marginal analysis in order to receive the point.

0	1 ✓

The response identifies the profit-maximizing output as 4 hats or does not explain that when the firm produces 4 hats, its marginal revenue $\text{D=MR}\ (\text{MR})$ is \$9 and its marginal cost (MC) is $\$8 (= \$32 - \$24)$. If the firm produces the fifth hat, its MR remains at \$9 but its MC rises to $\$10 (= \$42 - 32)$. A profit-maximizing firm will never produce at the quantity where MC exceeds MR .

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response must both state that the profit-maximizing output will remain as 4 hats and explain why in order to earn the point.

0	1 ✓

The response states that the profit-maximizing output remains 4 hats and explains that when the firm produces 4 hats, its marginal revenue MR is \$9 and its marginal cost (MC) is \$8. The rent is a fixed cost and does not change MC or MR ; therefore, there is no short-run impact on output.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

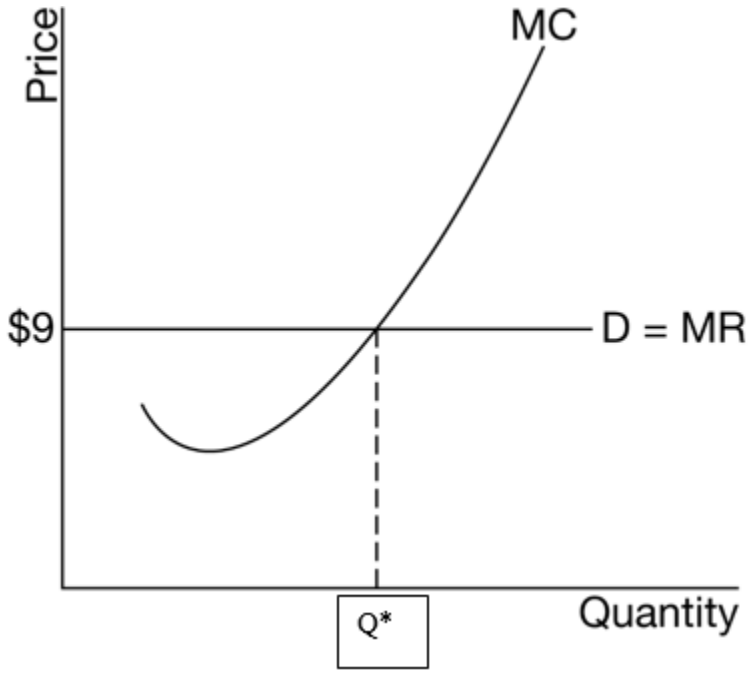
0	1	2 ✓

The response accurately includes both of the criteria below.

- ☐ A correctly labeled graph of the firm with a horizontal demand curve labeled $D = MR$ at the market price \$9

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- A rising marginal cost (MC) curve and the profit-maximizing quantity of hats as Q* at $P = MR = MC = \$9$



Include correctly labeled diagrams, if useful or required, in explaining your answers. A correctly labeled diagram must have all axes and curves clearly labeled and must show directional changes. If the question prompts you to "Calculate," you must show how you arrived at your final answer.

Assume that apples are produced in a perfectly competitive market. Grande's Orchard is a typical firm that grows and sells apples. Currently, Grande earns zero economic profit, and the market price of apples is $10 per bushel.

(a) Draw a correctly labeled graph showing Grande's demand curve, average total cost curve, and marginal cost curve, and show the profit-maximizing quantity, labeled Q_G .

(b) Suppose an increase in the popularity of apple cider increases the demand for apples. How will the increase in the demand for apples affect Grande's economic profit in the short run? Explain.

(c) What will happen to Grande's economic profit in the long run? Explain.

2. Respond to all parts of the question.

Part (a)

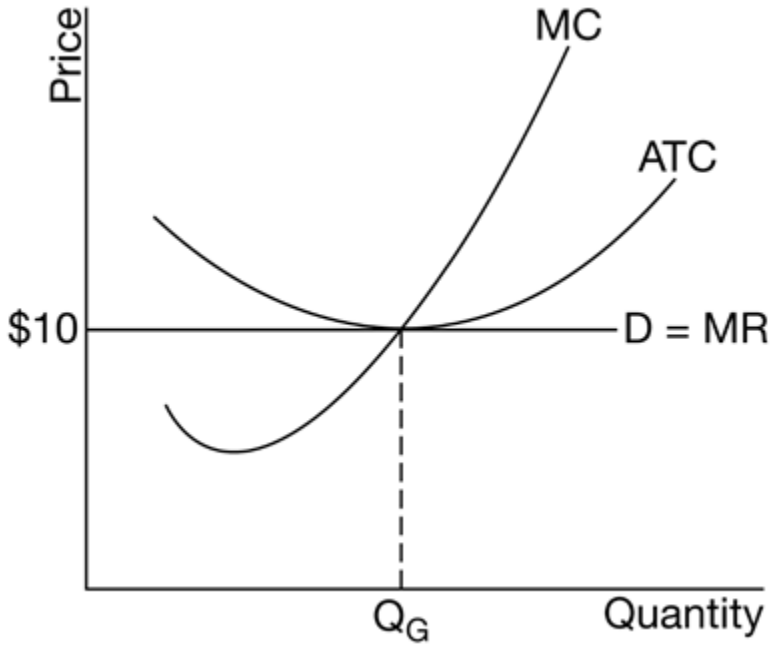
Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1	2	3 ✓

The response accurately includes three of the criteria below.

- ☐ A correctly labeled graph for the firm with a horizontal demand or marginal revenue (MR) curve at \$10
- ☐ A rising marginal cost (MC) curve and the profit-maximizing quantity, $\mathrm{Q_G}$, at $\mathrm{MR} = \mathrm{MC}$ on the graph
- ☐ The ATC curve is tangent to Grande's demand curve and the MC curve is passing through the minimum of the ATC curve



Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response must both state that Grande's profit will increase and explain why in order to receive the point.

0	1 ✓

The response states that Grande's profit will increase and explains that the increase in demand will raise the price of apples above Grande's average total cost, allowing Grande to earn short-run positive economic profit.

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Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The response must both state that Grande's profit in the long run will fall to zero and explain why in order to receive the point.

0	1 ✓

The response states that Grande's profit in the long run will fall to zero and explains that new firms will enter the market and increase the production of apples in the next growing season, causing the price of apples to fall to the minimum of average total cost.